

AYA YASSER AHMED

AI Engineer | Machine Learning & Computer Vision

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SUMMARY

AI Engineer specializing in **machine learning, computer vision, and generative AI**, with a strong track record of designing and deploying intelligent systems. Experienced in building **end-to-end ML pipelines**, optimizing deep learning models (e.g. reducing object detection error rates by 40%), and developing **GenAI-powered applications** using **LangChain, LangGraph, and IBM watsonx**. Skilled in **Python, TensorFlow, Pandas, and Pydantic** for data validation, preprocessing, and scalable analysis. Hands-on experience creating **AI assistants** for travel, claims analysis, and talent acquisition, as well as applying **OCR and computer vision** for document and image processing. Combines **technical rigor, problem-solving mindset, and UI/UX knowledge** to deliver **scalable, real-world AI solutions** that drive measurable impact.

EXPERIENCE

Front-End Development Intern, Route Academy An educational platform focusing on tech skills development Designed and implemented dynamic, user-centric web applications utilizing React.js and advanced JavaScript frameworks to deliver seamless performance and functionality.	08/2024 – 02/2025 Cairo, Egypt
IoT and ML Project Intern, Elsewedy Electric A leading company in electrical solutions and industrial automation Collected, analyzed, and visualized IoT data to improve operational efficiency.	06/2023 – 08/2023 Cairo, Egypt
AI and Computer Vision Trainee, Huawei A global leader in ICT infrastructure and smart devices Designed and trained a CNN-based object detector TensorFlow.	01/2023 – 02/2023 Cairo, Egypt
AI Engineer, Xyris - Built an AI-powered travel assistant chatbot (IBM watsonx & LangChain) that collects user info, validates completeness, and suggests best travel and hotel options. - Developed a Claims Data Analyzer capable of handling Excel files with inconsistent schemas using Pydantic for validation, Pandas for preprocessing, and anomaly detection for fraud/red flag identification. - Designed a Talent Acquisition AI (LangGraph) to automate recruitment by filtering, ranking, and interviewing candidates, including recording and rating performance. - Completed an AI Internship at Xyris Digital Solutions, gaining practical exposure to Generative AI, LLM-powered applications, and workflow automation. - Applied computer vision and OCR techniques (EasyOCR, PaddleOCR, OpenCV) in multilingual document analysis projects.	07/2025 – 09/2025 Cairo, Egypt

EDUCATION

Level 4, AI Department, Egyptian Russian University	09/2022 – 07/2026 Cairo, Egypt
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SKILLS

- Python
- Computer Vision
- JavaScript
- Responsive Web
- PaddleOCR
- LangGraph
- Prompt Engineering
- GitHub
- Figma
- React
- EasyOCR
- LangChain
- IBM watsonx Orchestrate

LANGUAGES

English

Advanced

PROJECTS

- | | |
|---|-------------------|
| AI-Powered Object Detection,
<i>A computer vision project aimed at object detection enhancement</i> <ul style="list-style-type: none">• Developed a basic object detection system using computer vision techniques. | 02/2025 – 04/2025 |
| Responsive E-Commerce Platform,
<i>An e-commerce platform project with a focus on responsiveness and user experience</i> <ul style="list-style-type: none">• Developed a production-ready e-commerce platform using React.js, Redux, and Next.js. | 05/2023 – 06/2023 |
| Brain Tumor Detection from X-rays <ul style="list-style-type: none">• Developed a CNN-based model (TensorFlow/Keras) to detect dementia indicators from brain X-rays, achieving 92% accuracy. | 01/2023 – 05/2023 |
| Travel Assistant Chatbot (IBM watsonx Orchestrate, LangChain) <ul style="list-style-type: none">• Built an AI-powered travel assistant that collects user trip details, validates missing information, and provides optimized travel options.• Integrated Amadeus API to fetch flight offers with the best prices, then suggested hotel options by category.• Implemented with both IBM Watsonx Orchestrate tools and LangChain agent workflows to compare approaches. | 07/2025 |
| Claims Data Analyzer (Python, Pandas, Streamlit) <ul style="list-style-type: none">• Developed a claims analysis tool capable of processing Excel files with varying column names by mapping them to a predefined schema.• Automated data cleaning, normalization, and analysis to detect anomalies and red flags exceeding threshold limits.• Delivered results through interactive dashboards, enabling quick identification of fraudulent or suspicious claims. | 07/2025 – 07/2025 |
| AI-Powered Talent Acquisition System (LangGraph, LLMs) <ul style="list-style-type: none">• Designed a recruitment assistant to streamline candidate selection for vacancies.• Processed uploaded CVs, filtered, and ranked candidates based on job description matching.• Managed the interview pipeline, including scheduling, recording interviews, and automatically rating candidate performance. | 08/2025 – 09/2025 |

KEY ACHIEVEMENTS

AI & Computer Vision, *Brain Tumor Detection from X-rays*

- Developed a CNN-based model (TensorFlow/Keras) to detect dementia indicators from brain X-rays, achieving 92% accuracy.

AI & Computer Vision, *Created an AI model that analyzes images to determine health status, identifying diseases when the subject is not healthy.*

E-Commerce Platform Development,

Developed a responsive React.js storefront with dynamic product recommendations.

UI/UX with AI Integration

- Shows full product lifecycle (Figma React Native A/B testing).